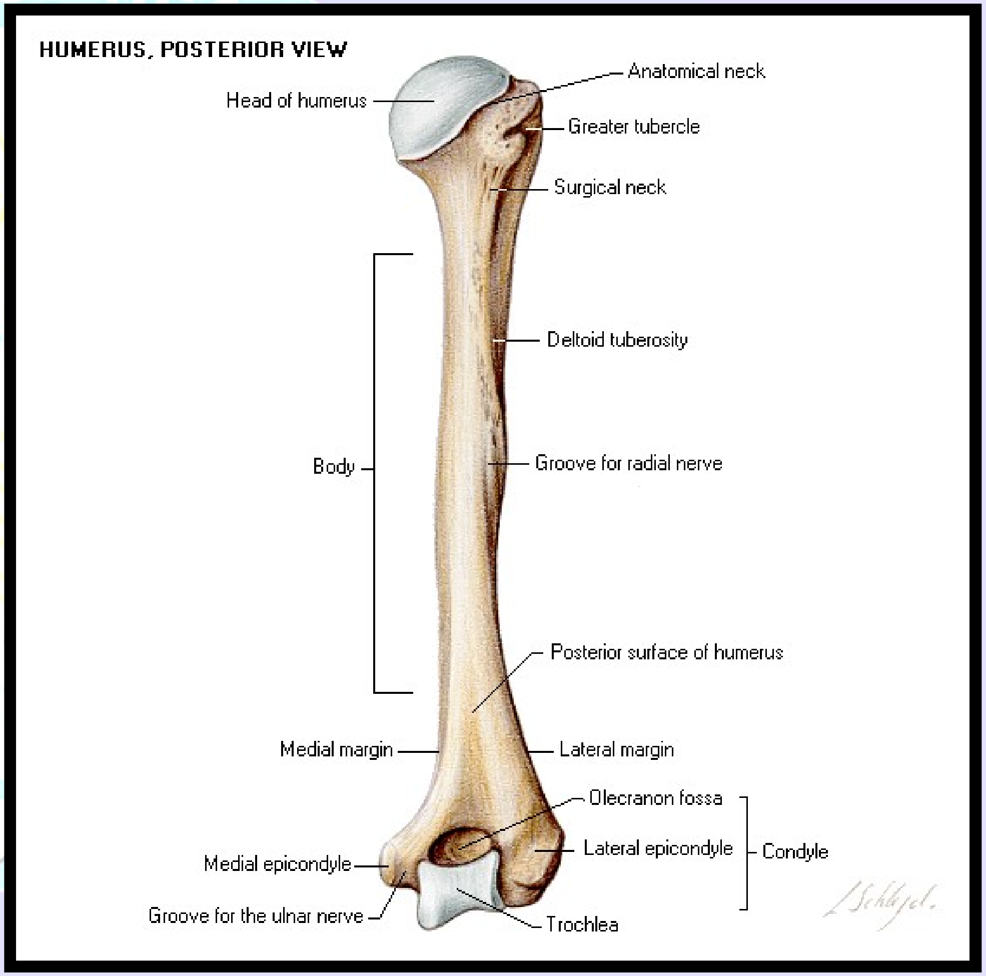
CLASSIFICATION
&
HISTOLOGY
OF
BONE

LEARNING OBJECTIVES

- At the end of the lecture the student should be able to:
  - Recognize bone and its functions and composition.
  - Differentiate between woven bone and lamellar bone.
  - Differentiate between compact bone and spongy bone.
  - Learn the applied aspect of bone

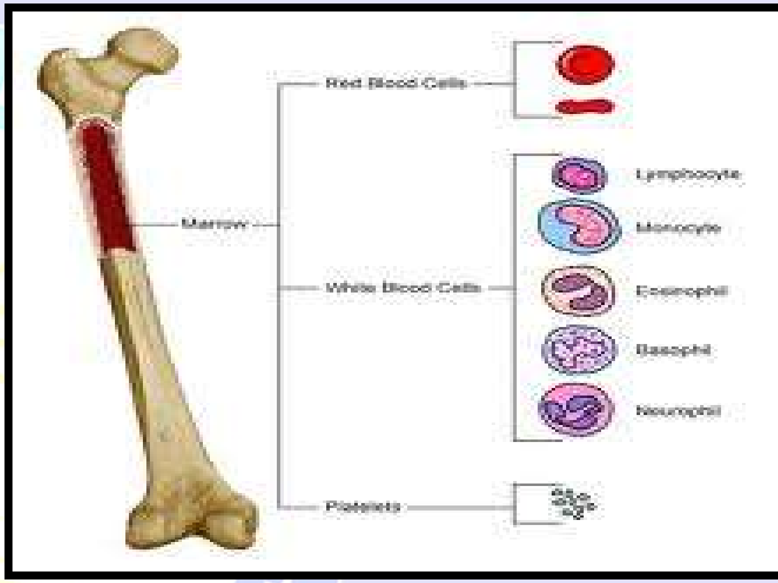
BONE

- **Bone** is a specialized type of connective tissue characterized by intercellular calcified material - **The bone matrix**



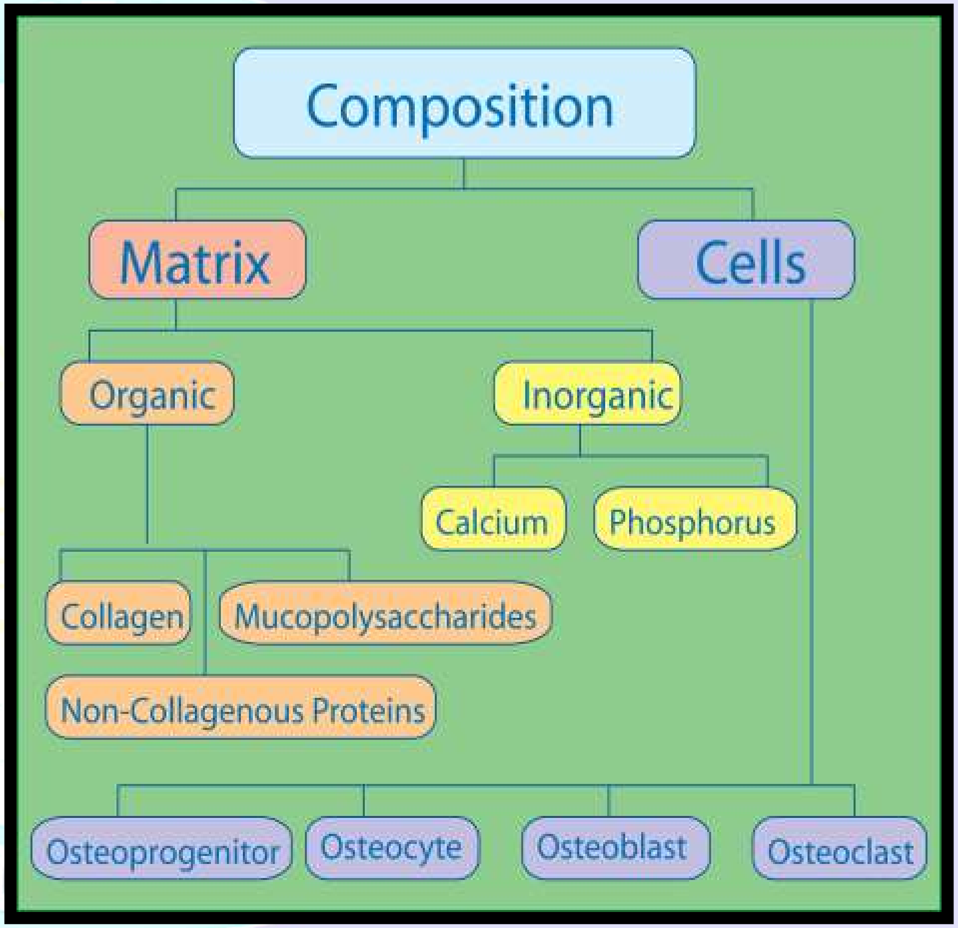
FUNCTIONS OF BONE

- Supports soft tissue
- Protects vital organs (cranium, thoracic cavity)
- Contains bone marrow
- Reservoir of Ca, PO4 and maintains their constant concentrations in body fluids
- Locomotion

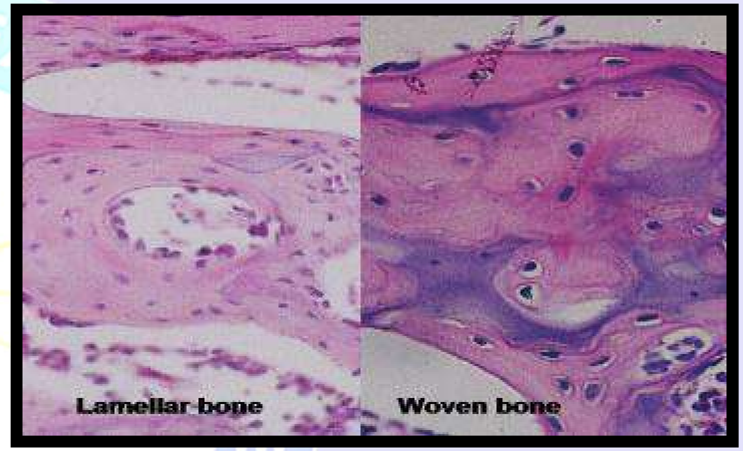


COMPOSITION

- **Cells**
  - Osteoblasts
  - Osteocytes
  - Osteoclasts
- **Bone matrix**
  - Organic:
    Fiber + ground substance
    (Proteoglycan and glycoprotein)
  - Inorganic:
    Calcium + phosphorus (hydroxyapatite crystal & noncrystalline amorphous form), bicarbonates and citrate etc



TYPES OF BONE



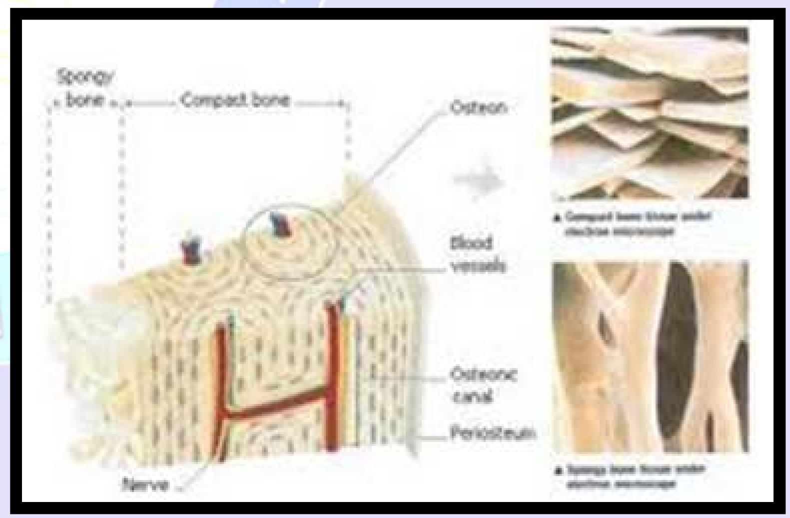
Microscopically

- Primary (woven)
- Secondary (Lamellar)

Macroscopically

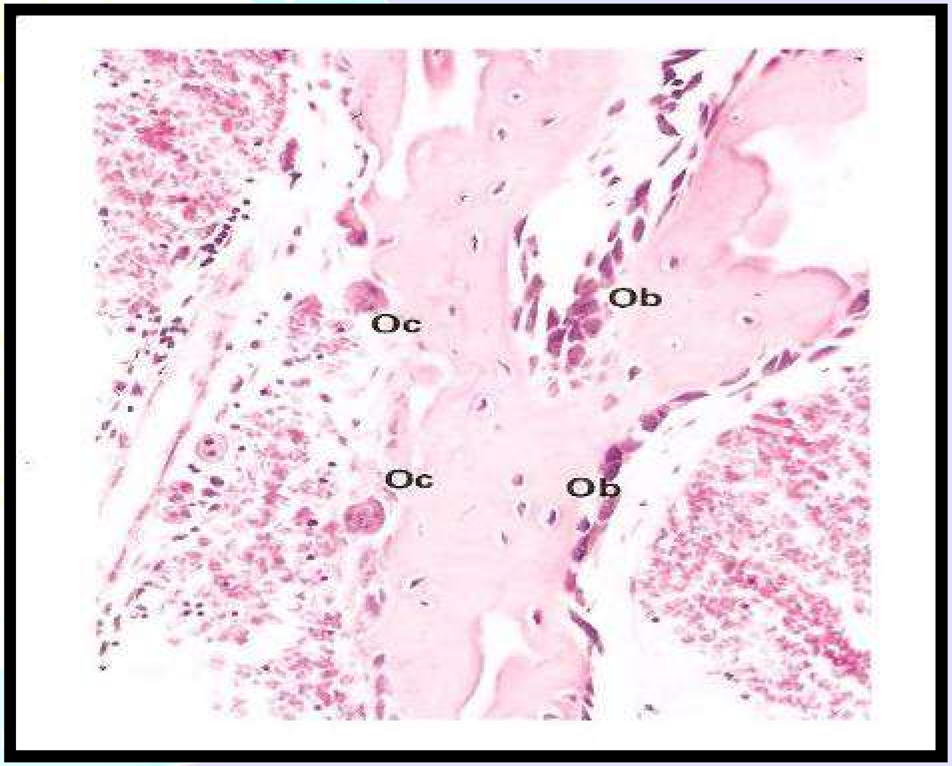
- Compact
- Cancellous

(Both have lamellar arrangement)



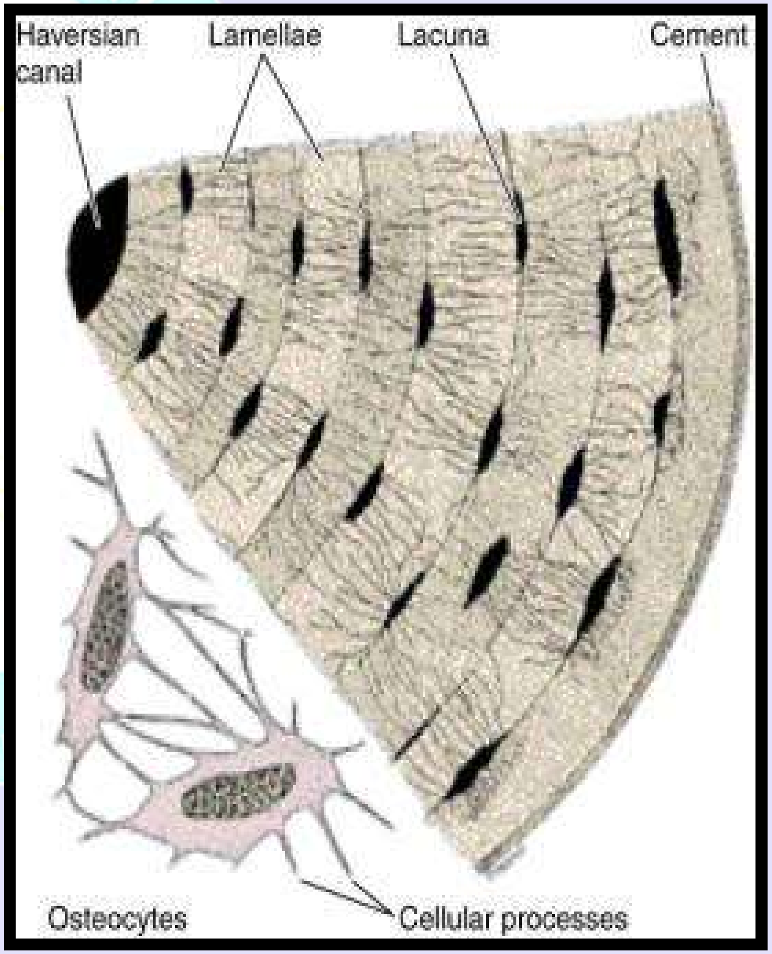
WOVEN BONE

- Characterized by random deposition of fine collagen and increased cellularity.
- Relatively radiolucent (low mineral contents).
- Seen in embryonic development, fracture and repair.



LAMELLAR BONE

- Characterized by
- Arrangement of collagen fibers in lamellae organized concentrically around a vascular canal containing blood vessels and nerves.
- The lamellae and the vascular canal constitute the Haversian system or osteon.
- Lacunae containing osteocytes are present between the lamellae.
- Mineralized matrix with few fibers present around each osteon, called the cementing substance.



DIFFERENCE BETWEEN WOVEN AND LAMELLAR BONES

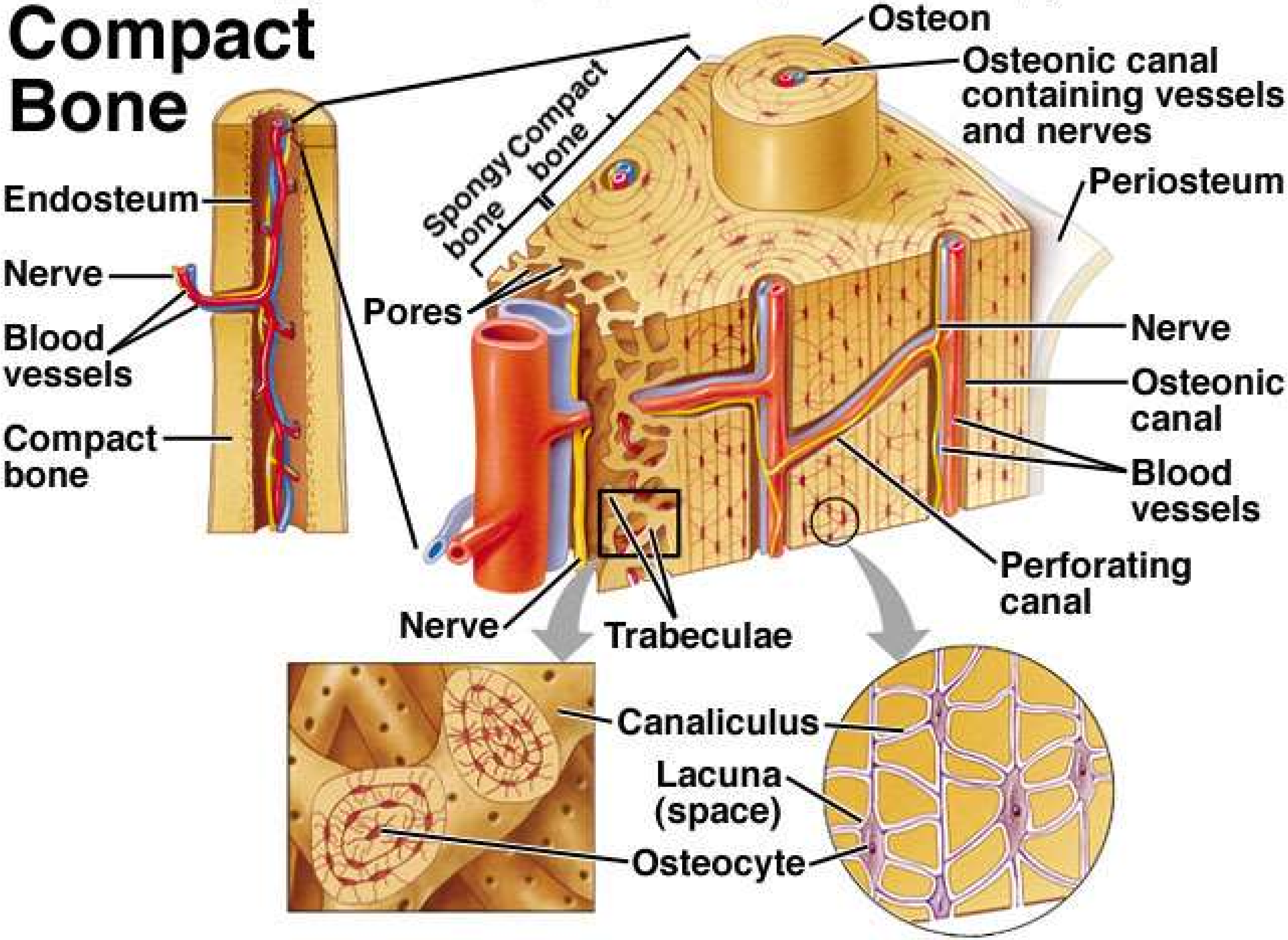
- Woven (primary) bone tissue is temporary and is replaced in adults by Lamellar (secondary) bone tissue except in few places i.e. near sutures of flat bones of skull
- There is irregular array of collagen fibers in Woven (primary) bone tissue
- Woven (primary) bone tissue has lower mineral content
- Woven (primary) bone tissue has higher proportion of osteocytes

COMPACT BONE

- 4 Types of lamellae seen:
  - a. Haversian system
  - b. Outer circumferential
  - c. Inner circumferential
  - d. Interstitial

- Volkman’s canal communicate the osteons and, periosteal and endosteal surfaces

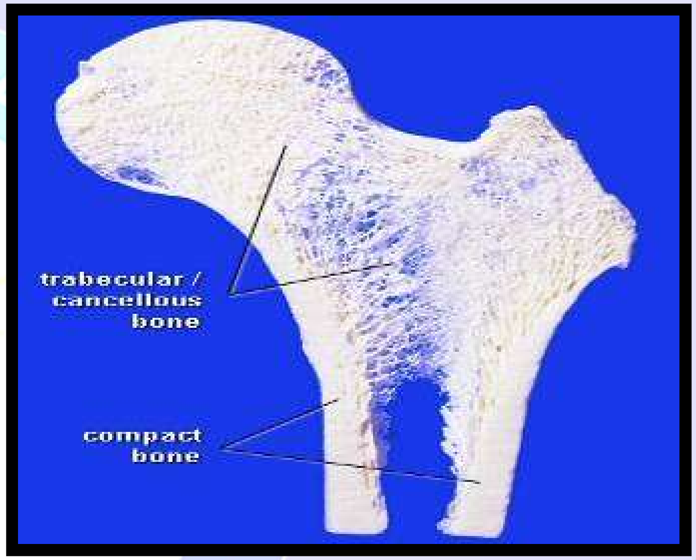
Compact Bone

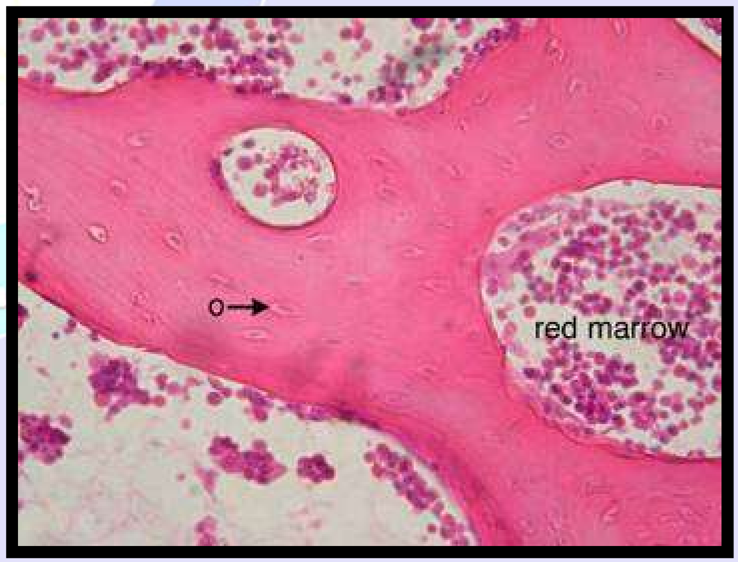


SPONGY BONE

- Bony tissue is in the form of trabeculae having lamellar arrangement.

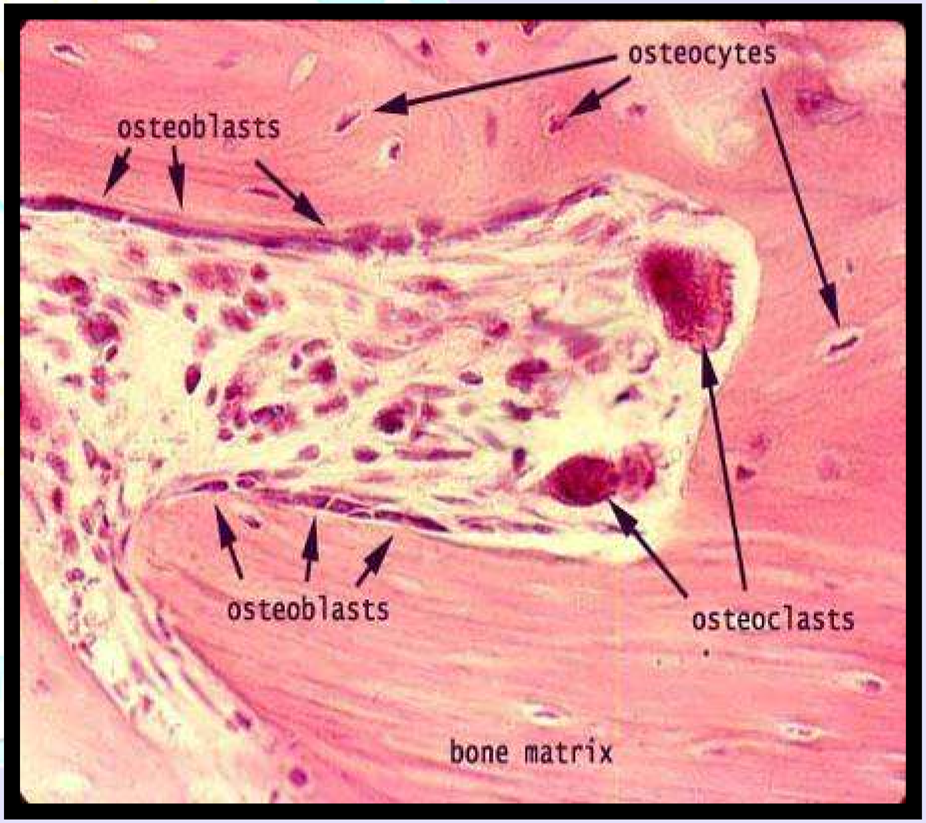
- Vascular canals are absent as the vessels in marrow cavities provide nutrition by means of diffusion.





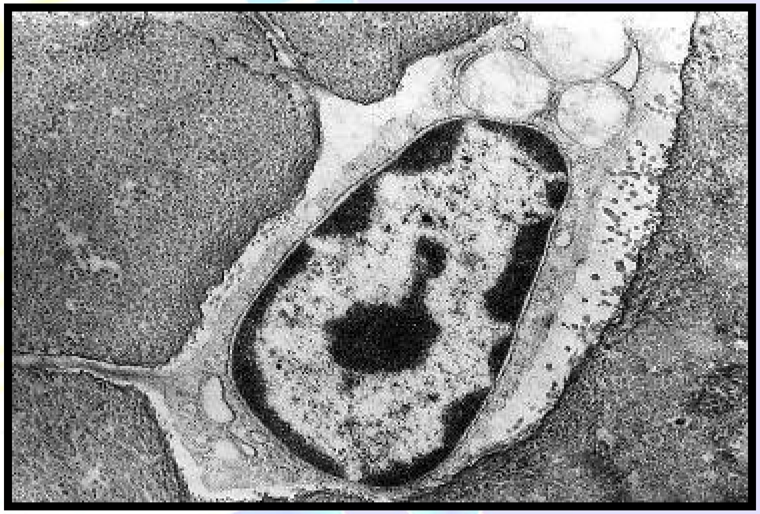
OSTEOBLASTS

- Synthesize organic components of matrix (collagen type 1, proteoglycans, glycoproteins.)
- Epitheloid appearance, cuboidal or columnar shape
- Shows alkaline phosphatase activity
- Influence deposit of Ca++, PO4.
- Estrogen, PTH stimulate its activity



OSTEOCYTES

- Mature bone cells that sit in lacunae
- Almond shaped cell with condensed nuclear chromatin
- Gap junctions between osteocytes provide nutrition Maintain bony matrix; death causes resorption of matrix
- Stimulated by calcitonin; inhibited by PTH

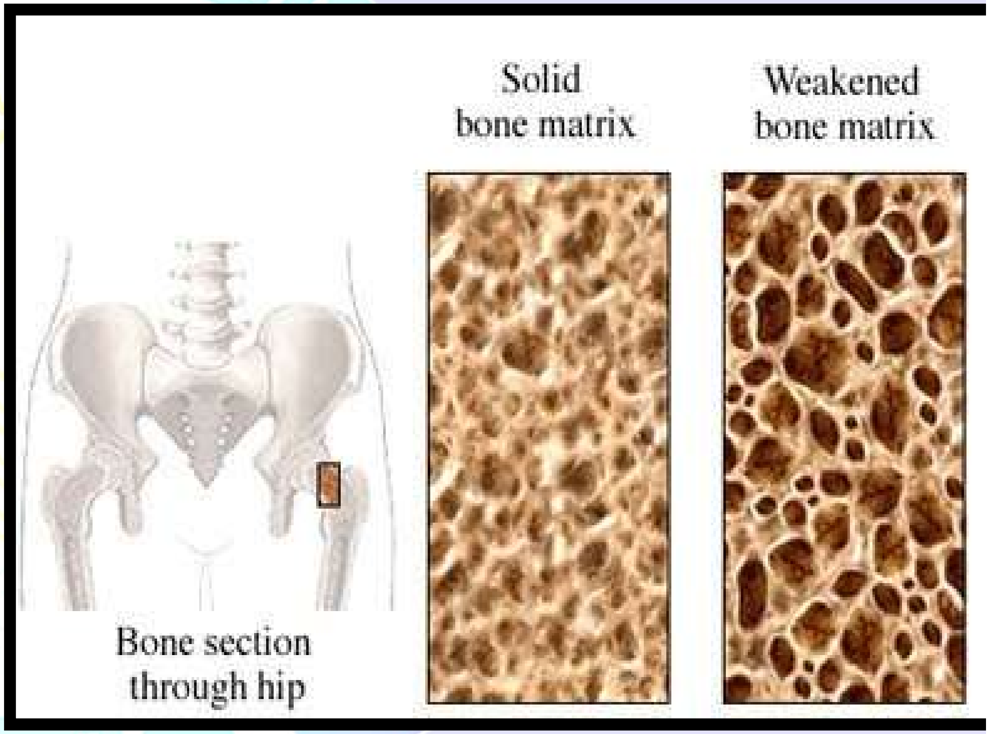


OSTEOCLASTS

- Derived from monocytes; engulf bony material
- Large, multinucleated and branched found in Howship's lacunae
- Surface foldings forming ruffled border increasing the active resorptive area.
- Secrete enzymes (colleganase etc) that digest matrix

BONE MATRIX

- Organic – 35% collagen
  - Chondroitin sulfate, hyaluronic acid and osteocalcin
- Inorganic – 65% calcium phosphate minerals – hydroxyapatite crystals, which binds water around it. This hydration shell facilitate exchange of ions between it and body fluids.
- Combination provides strength and rigidity. Laid down by osteoblasts.

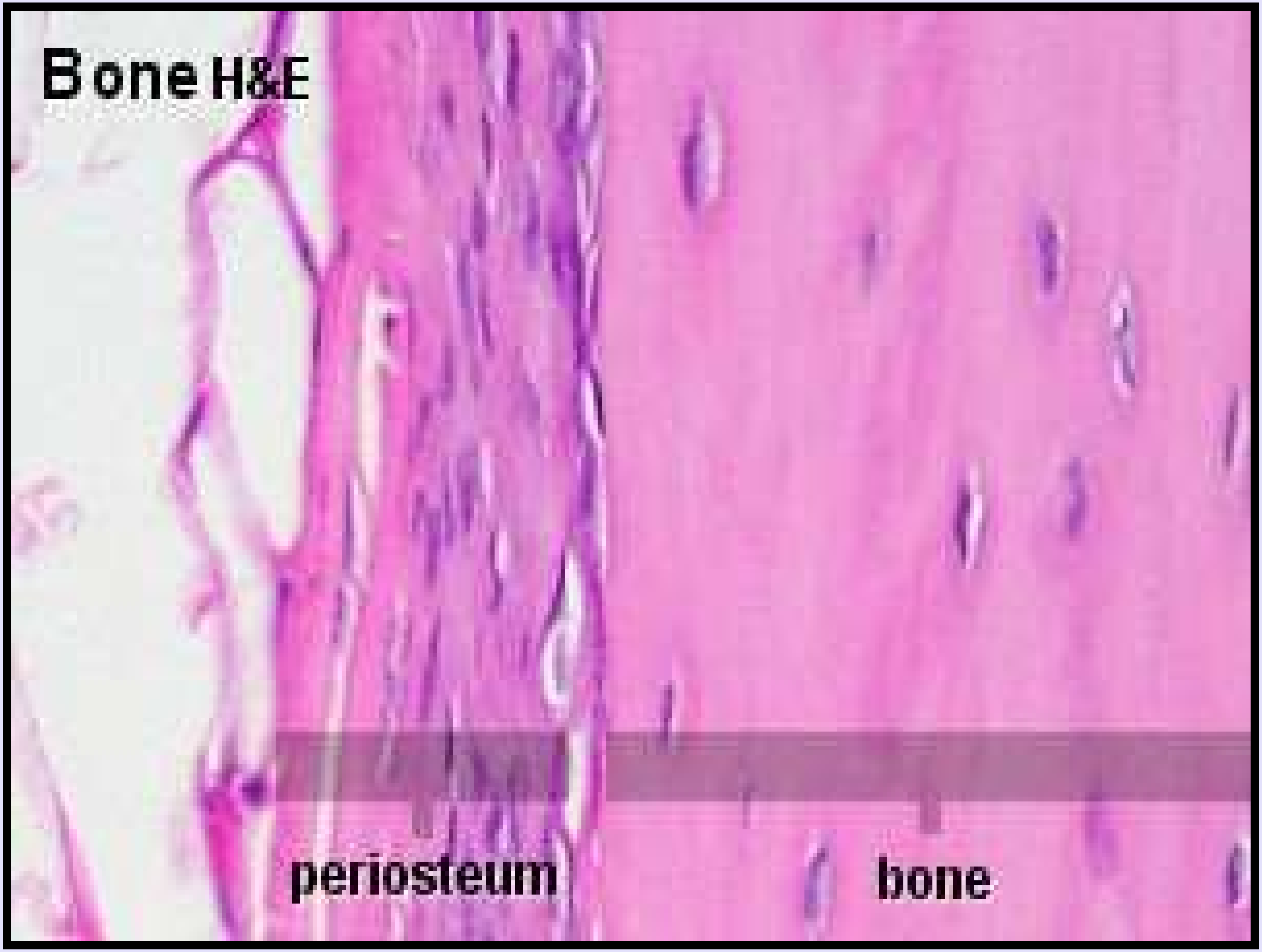


PERIOSTEUM

- Consists of outer layer of collagen fibers and fibroblast. The fibers penetrate the bone matrix the **Sharpey's fiber**
- Inner layer of reserve cells which can differentiate in osteoblast cells

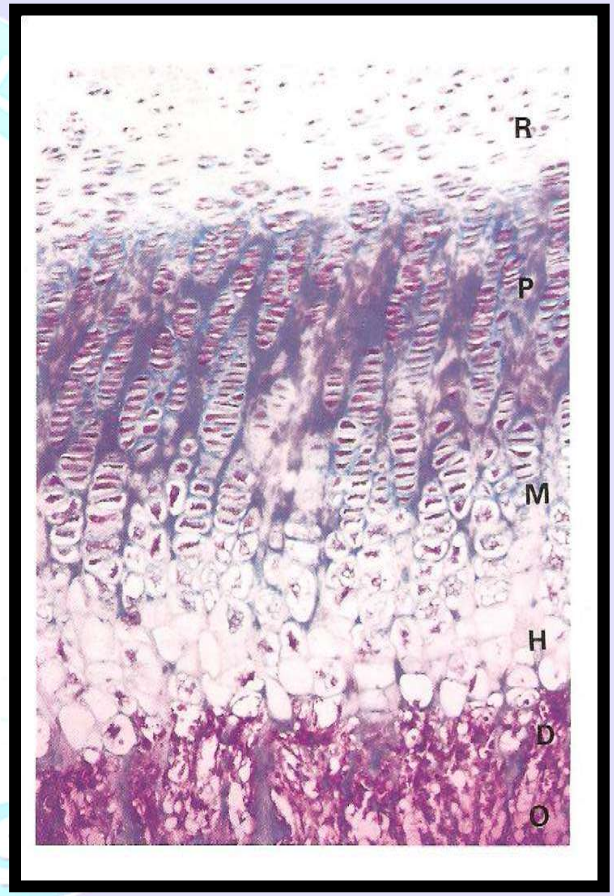
ENDOSTEUM

- Very little connective tissue and single layer of reserve cells
- Lines medullary and marrow cavities of bone


Bone H&E
periosteum
bone

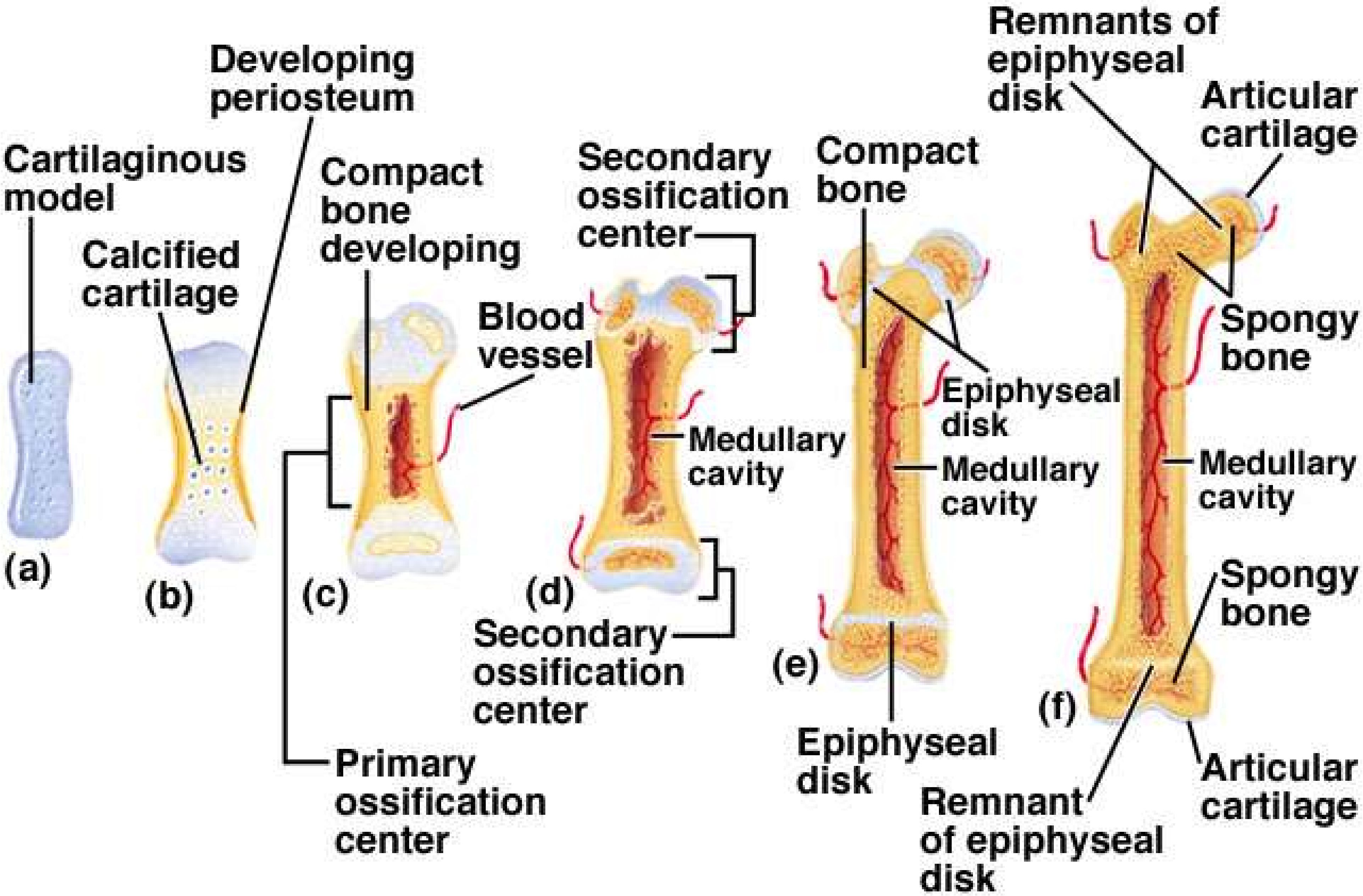
ENDOCHONDRIAL BONE FORMATION

- Within the model of hyaline cartilage.
- Hypertrophy and destruction of chondrocytes.
- Calcification of cartilage matrix.
- Arrival of osteogenic bud.
- Osteoblast differentiate from progenetor cells and form bone matrix.



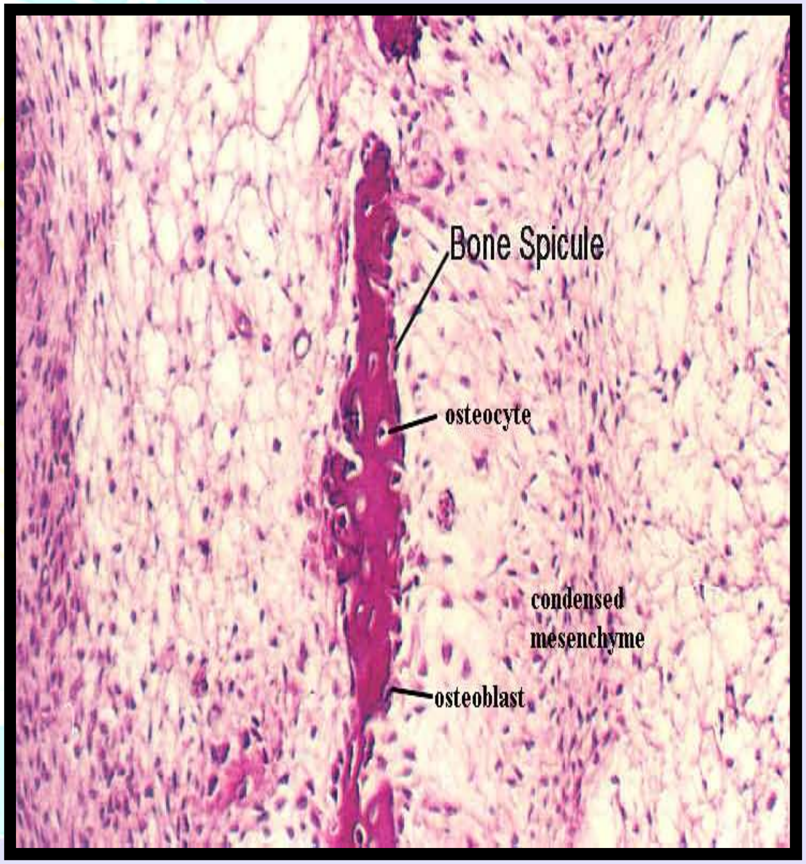
Epiphysial growth plate

Endochondral Bone Development

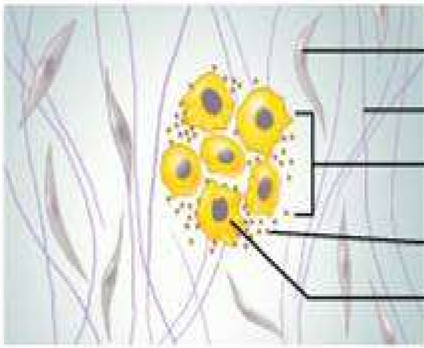


INTRAMEMBRANOUS BONE FORMATION

- Connective tissue develops embryonically and gives rise to bone.
- Cells in connective tissue differentiate into osteoblasts.
- Osteoblasts lay down collagen eventually starts to calcify and become osteocytes.
- Cancellous bone eventually becomes compact bone.

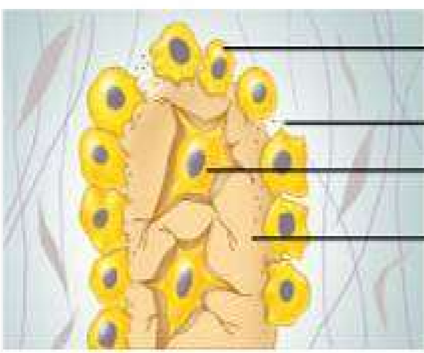


INTRAMEMBRANOUS OSSIFICATION



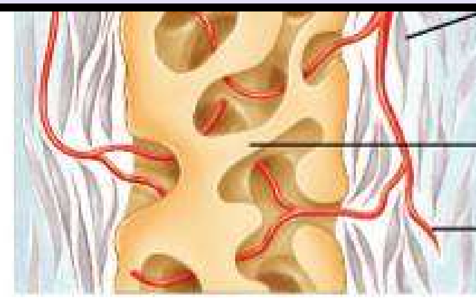
① **An ossification center appears in the fibrous connective tissue membrane.**

- Selected centrally located mesenchymal cells cluster and differentiate into osteoblasts, forming an ossification center.



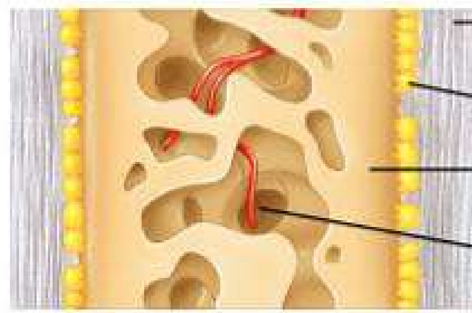
② **Bone matrix (osteoid) is secreted within the fibrous membrane.**

- Osteoblasts begin to secrete osteoid, which is mineralized within a few days.
- Trapped osteoblasts become osteocytes.



③ **Woven bone and periosteum form.**

- Accumulating osteoid is laid down between embryonic blood vessels, which form a random network. The result is a network (instead of lamellae) of trabeculae.
- Vascularized mesenchyme condenses on the external face of the woven bone and becomes the periosteum.



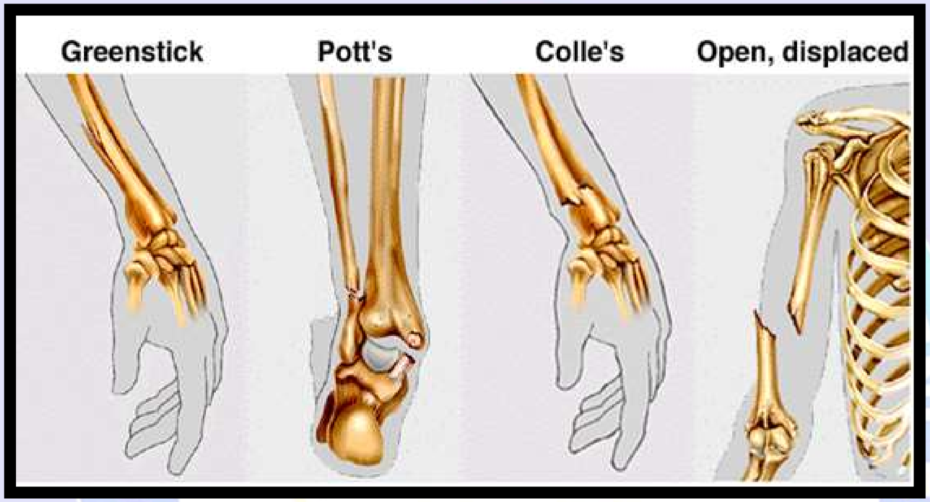
④ **Bone collar of compact bone forms and red marrow appears.**

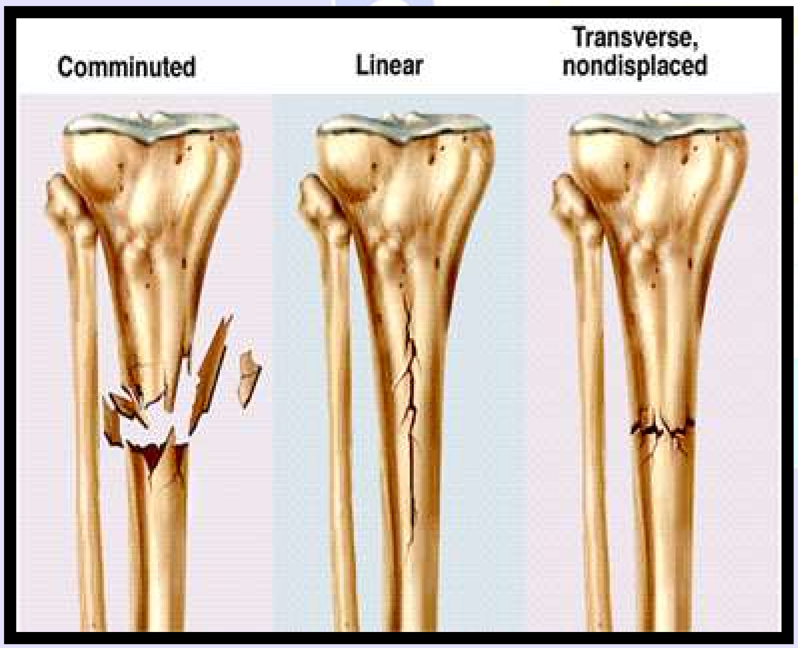
- Trabeculae just deep to the periosteum thicken, forming a woven bone collar that is later replaced with mature lamellar bone.
- Spongy bone (diploë), consisting of distinct trabeculae, persists internally and its vascular tissue becomes red marrow.

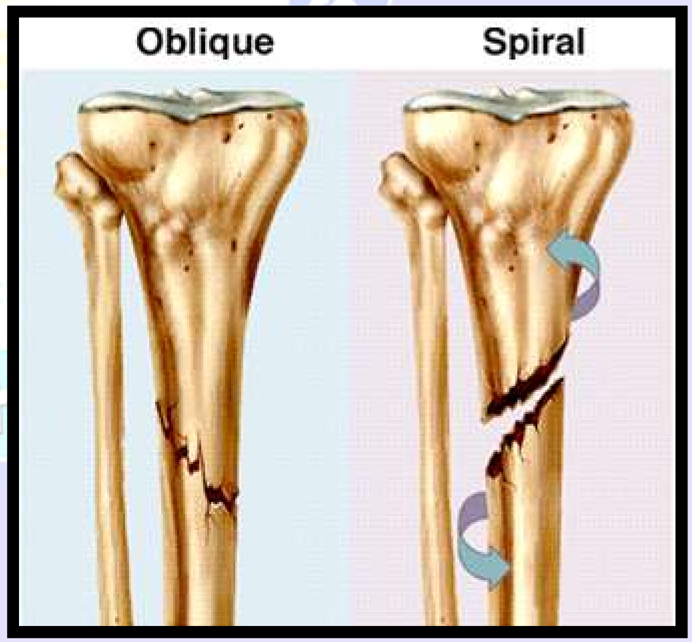
APPLIED ANATOMY

- **Fractures**
  - Open and
  - Closed
- **Healing of these takes place in three stages;**
  - repair by granulation tissue
  - union by callus
  - consolidation by mature bone

TYPES OF FRACTURES

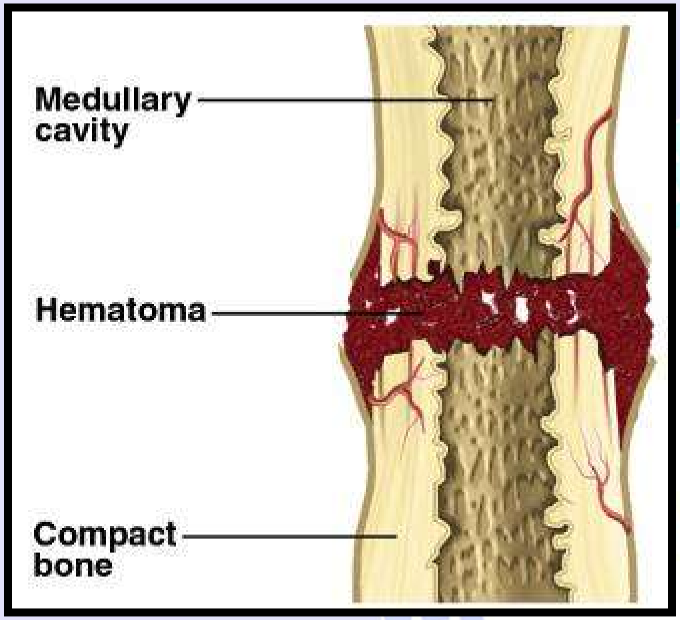






HEALING OF FRACTURES

1



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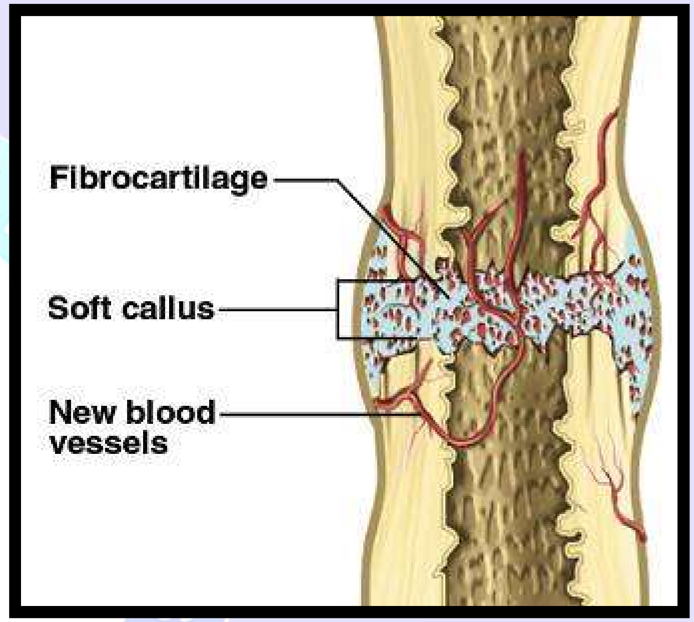

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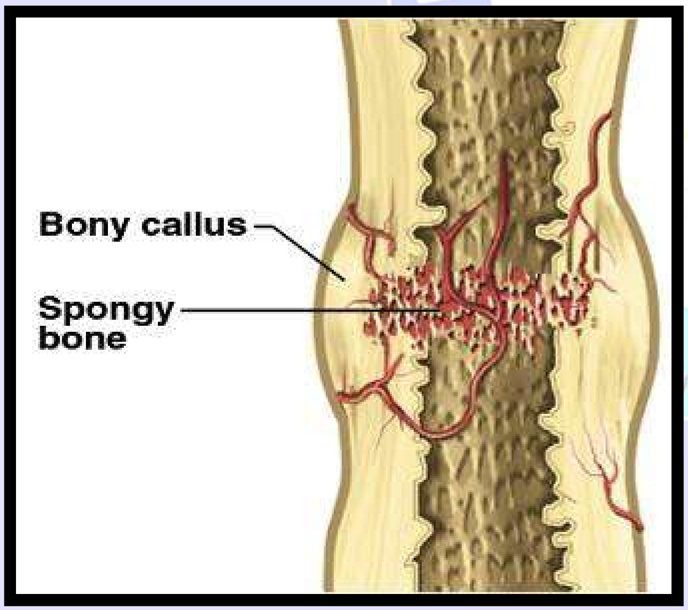

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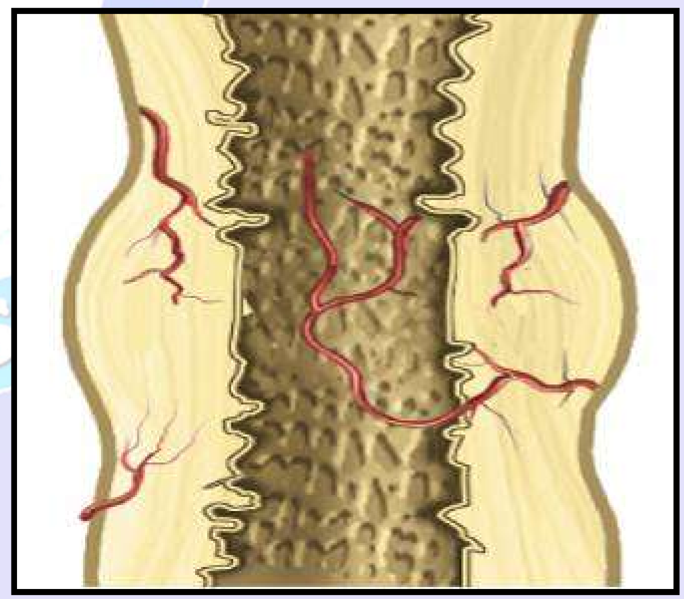
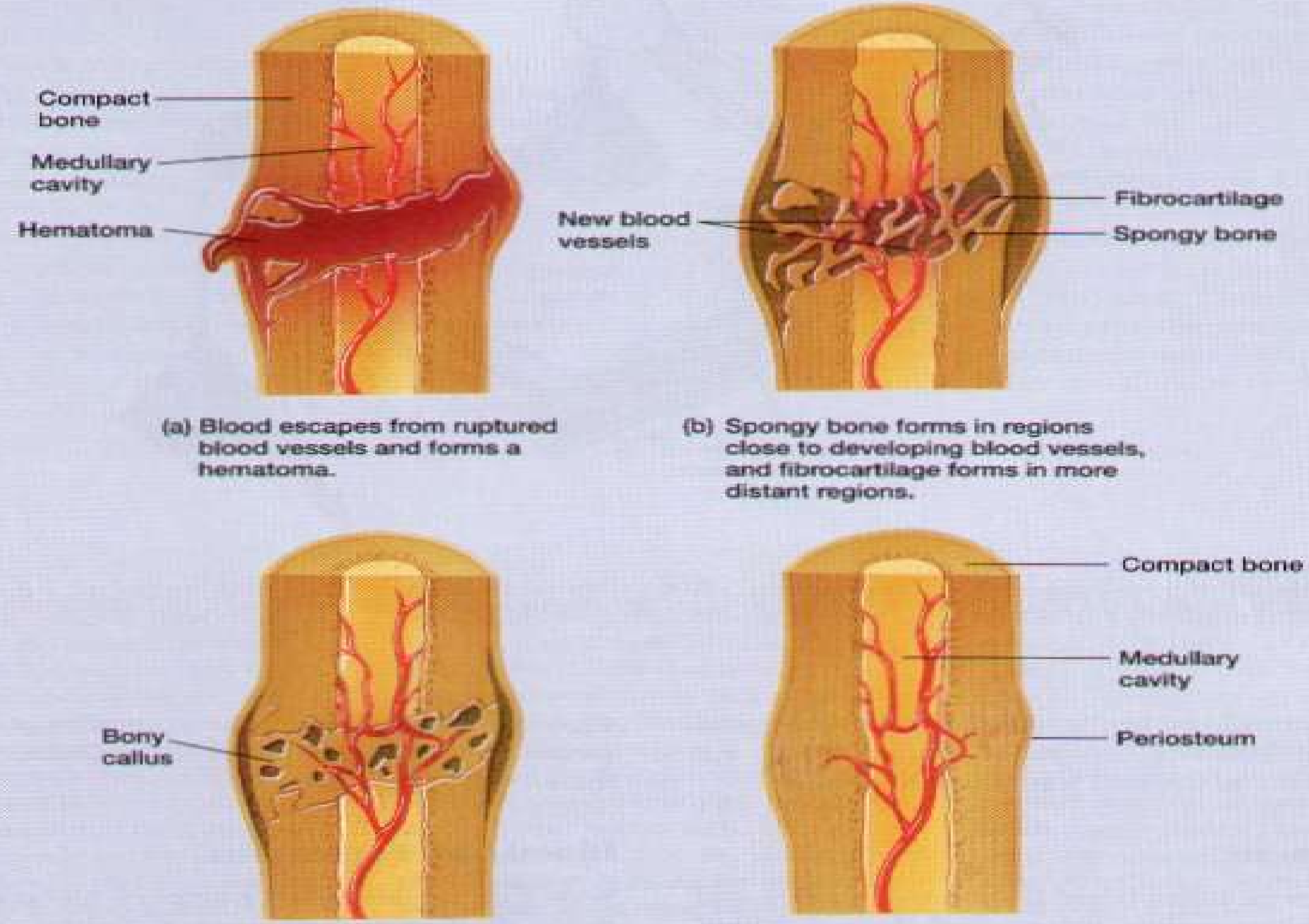


FRACTURE REPAIR



(a) Blood escapes from ruptured blood vessels and forms a hematoma.

(b) Spongy bone forms in regions close to developing blood vessels, and fibrocartilage forms in more distant regions.

(c) A bony callus replaces fibrocartilage.

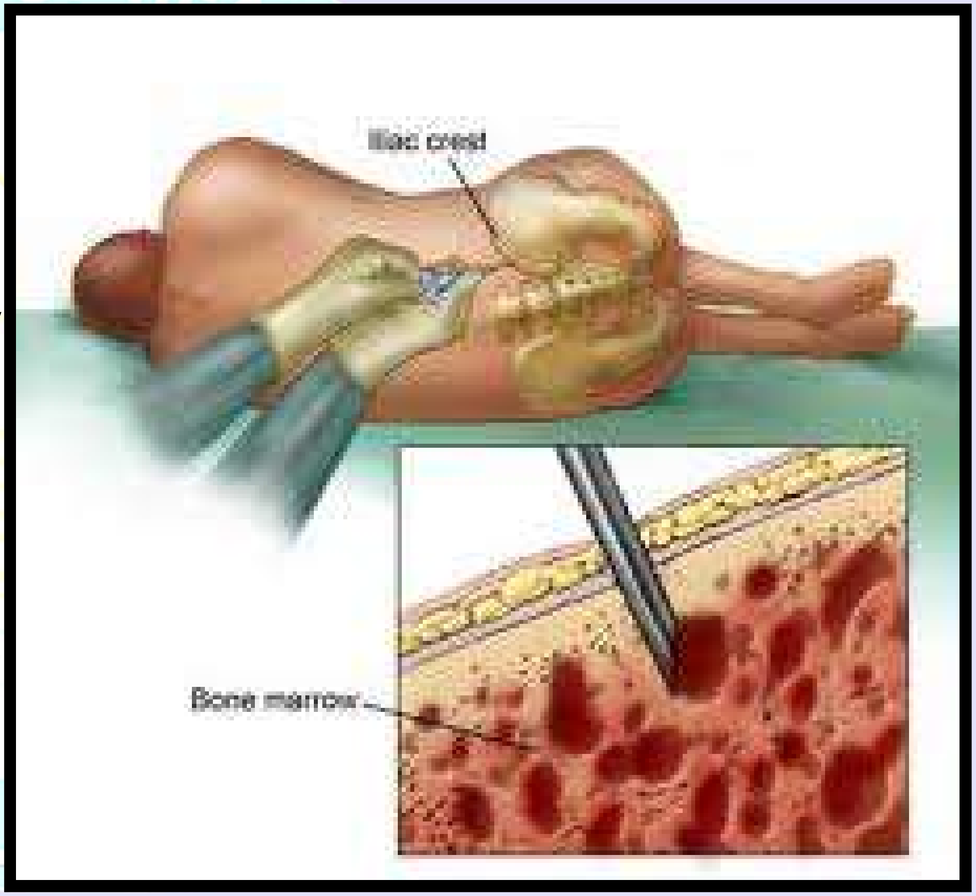
(d) Osteoclasts remove excess bony tissue, restoring new bone structure much like the original.

OSTEOPOROSIS

- Most common bone disease
- Bones lose mass & becomes brittle due to loss of both organic matrix & minerals
  - Risk of fracture of hip, wrist & vertebral column.
  - Leads to fatal complications such as pneumonia or pulmonary embolism.
  - Widow’s (dowager’s) hump is deformed spine
  - Postmenopausal white women at greatest risk
    - By age 70, average loss is 30% of bone mass
    - ERT slows its progress, but best treatment is prevention -- exercise & calcium intake (1000 mg/day) between ages 25 and 40.

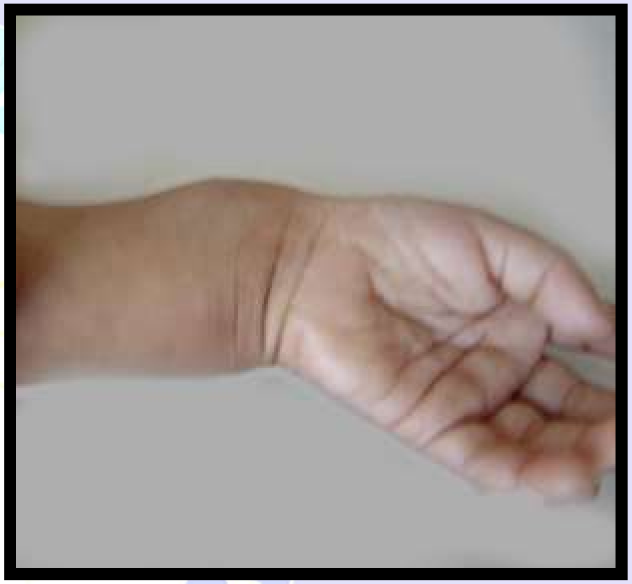
BONE MAROW BIOPSY

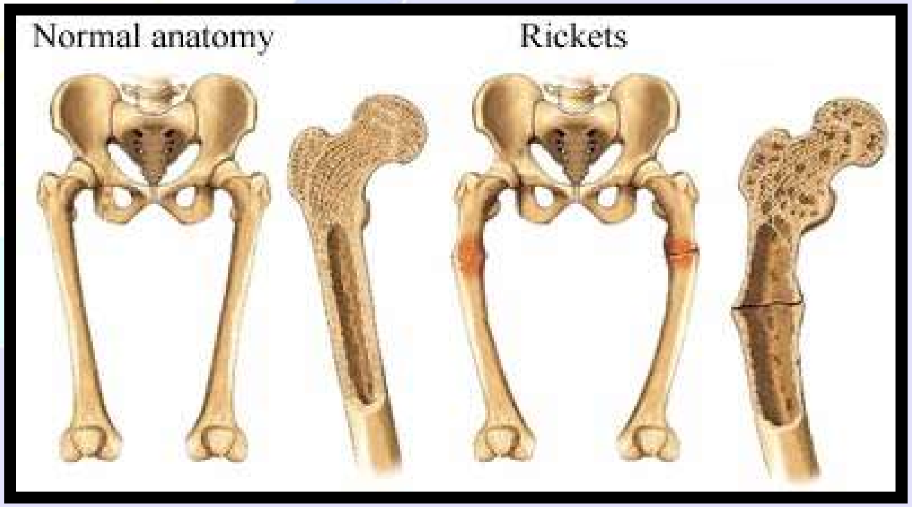
- Passing a metal pin into the medullary cavity, during bone marrow biopsy hardly interferes with the blood supply of the bone, as it is richly supplied.



RICKETS

- Calcification of cartilage fails and ossification of the growth zone is disturbed.
- Osteoid tissue is formed normally and the cartilage cells proliferate freely, but mineralization does not take place.
- This results in craniotabs, rachitic rosary, enlarged epiphyses in limb bones, and spinal & pelvic deformities.
  -Fractures, tumors, and infections of the bone are very painful.





REFERENCES

- MEDICAL HISTOLOGY BY LAIQ HUSSAIN SIDDIQUI
- BASIC HISTOLOGY BY JUNQUEIRA

THANK YOU!